Tutorial (1)

Student Name: **ID:** **Section:**

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Question 1: Classify the following variables as Qualitative or Quantitative.	**Answer**
1) The academic majors of undergraduate students in the Common First Year.	.....
2) The exact time (in seconds) recorded for a runner to finish a 400-meter race.	.....
3) The blood type of a patient admitted to a hospital.	.....
4) The monthly internet data usage of a mobile phone subscriber.	.....
Question 2: Classify the following variables as Discrete or Continuous.	**Answer**
1) The number of customer calls received by a service center between 8:00 AM and 9:00 AM.	.....
2) The exact weight of a randomly selected newborn baby.	.....
3) The capacity (in liters) of fuel tanks in passenger cars.	.....
4) The total number of defective items found in a batch of 500 manufactured products.	.....

Question 3: Select the correct answer for each of the following:

1) Numerical measures calculated from sample data are called

a) Parameter b) Statistic c) Qualitative variable d) Population

2) When constructing a frequency distribution, if the lower limit of the first class is 10 and the class width is C = 5, the upper boundary of the first class is

a) 14 b) 15 c) 14.5 d) 15.5

3) The appropriate graph used to represent continuous quantitative data divided into intervals is the

a) Bar Chart b) Pie Chart c) Histogram d) Multiple Bar Chart

4) In a pie chart, if the relative frequency of a category is 0.25, the central angle representing this sector is

a) 25^0 b) 45^0 c) 90^0 d) 180^0

Question 4: The following frequency distribution table presents the sleeping time of 100 individuals. Use the table to answer the questions that follow:

Class Boundaries	Frequency	Relative Frequency	Percent Frequency	ACF	DCF
1 - 3	15				
3 - 5	A			E	
5 - 7	30	C			F
7 - 9	15		D		
Total	B				

1. Complete the summary table below based on the above frequency distribution table.

Symbol	A	B	C	D	E	F
value	……	……..	……..	……..	……..	……..

2. What percentage of observations is greater than or equal 5 h?

……

3. What percentage of observations is less than 7 h?

……

4. Represent the above data using a histogram.



The End– Good Luck

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Question 5: Answer the following statements with True or False.		**Answer**
1	A discrete variable can take any uncountable real value within a given interval.	.....
2	A qualitative variable can take numerical values.	.....
3	Inferential statistics allows us to reach conclusions and make generalizations about an entire population based on sample data.	.....
4	A sample always contains more information than the population from which it was drawn.	.....
5	Student ID numbers are considered qualitative variables even though they consist of digits.	.....
6	In a histogram, the vertical bars must be separated by equal spaces.	.....
7	The cumulative frequency of the last class is always equal to the total sample size n.	.....
8	A frequency polygon is constructed by connecting points with coordinates (x_i, f_i) using straight lines, where x_i is the class midpoint.	.....

Question 6: Classify the following as Population or Sample.		**Answer**
1	The monthly salaries of all faculty members at King Saud University.	.....
2	A group of 45 blood samples chosen from a blood bank to test for a specific virus.	.....
3	The weights of 100 sheep selected from a fattening livestock farm to check average growth.	.....
4	All palm trees currently planted in the Qassim region.	.....

Question 7: Classify the following as Parameter or Statistic.		**Answer**
1	The maximum temperature recorded across all meteorological stations in a nation during July.	.....
2	The average GPA calculated for 35 graduating students selected from the College of Science.	.....
3	The proportion of students who passed an English placement test calculated from a group of 60 freshmen.	.....
4	The average of heights for all 18-year-old citizens in a country.	.....

Question 8: Consider a study of 300 males aged 18 years or higher conducted in a city called D. The study indicates that there are 10% smokers. Answer the following questions:

1. What is the sample?

.....

2. What is the population?

.....

3. What is the variable of interest?

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The End– Good Luck